10. (a) A binary signal $x_1(t) = 0$ for $t < 0$. For positive time, $x_1(t)$ toggles between one and zero in every one second. Determine the energy and power of $x_1(t)$.

$t=1,\ P=1/2$
$t=2,\ P=1/4$
$t=3,\ P=2/6$
$t=4,\ P=2/8$
$t=5,\ P=3/10$

$= \frac{\lceil t/2 \rceil}{2t}$

$$E = \int_0^1 1\,dt + \int_2^3 1\,dt + \cdots = \infty$$

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} (x(t))^2\,dt = \lim_{T\to\infty} \frac{\lceil t/2 \rceil}{2} = 1/4$$

(b) A binary signal $x_2(t) = 0$ for $t < 0$. For positive time, $x_2(t)$ toggles between one and zero as follows: one for 1 second, zero for 1 second, one for 1 second, zero for 2 seconds, one for 1 second, zero for 3 seconds, and so forth. That is, the "on" time is always one second, but the "off" time successively increases by one second between each toggle. A portion of $x_2(t)$ is shown below. Determine the energy and power of $x_2(t)$.

$t=1,\ P=1/2$
$t=2,\ P=1/4$
$t=3,\ P=2/6$
$t=4,\ P=2/8$
$t=5,\ P=2/10$
$t=6,\ P=3/12$



1,2 → 1
3,4,5 → 2
6,7,8,9 → 3
10,11,12,13,14 → 4

$$E = \int_0^1 1\,dt + \int_2^3 1\,dt + \int_5^6 dt + \cdots = \infty$$

$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} (x(t))^2\,dt = \lim_{t\to\infty} \frac{\left\lfloor \frac{-1/2 + \sqrt{1+8t}}{2} \right\rfloor}{2t} = \frac{\left\lfloor \frac{\sqrt{8t+1} - 1}{2} \right\rfloor}{2t} \to 0$$